

matchgroup_test_routed

Design Report

kicad-tools 0.13.0

Rev 1 | 2026-06-12 | jlcpb

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	8 (5 SMD, 3 THT, 0 other)
Nets	34
Traces	1150 segments
Vias	48
Board Size	110.0 x 95.0 mm

Design Overview

Theory of Operation

Match-Group Test Board

N-trace + group-of-pairs match-group regression testbench

Epic #2661 Phase 3L (issue #2724)

Power Architecture

Power Rails: PWR_FLAG

Assembly Notes

5 fine-pitch components

- **Fine-pitch components:** 5 (U2, U1, U5, U4, U3)

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: SKIPPED – ERC skipped by user request

Bill of Materials

Value	Package	Qty	References
ADDR_HDR	PinHeader_1x09_P2.54mm_Vertical	1	J3
FFC6_MIP1	FFC_6P_1.0mm	1	J1
HDMI19	HDMI_A_Receptacle	1	J2
BGA49_HDMI	BGA-49_5.0x5.0mm_Layout7x7_P0.5mm	1	U4
QFN24_MIP1	QFN-24-1EP_4x4mm_P0.5mm	1	U3
QFN48_DDR_CTRL	QFN-48-1EP_7x7mm_P0.5mm	1	U1
QFN48_DRAM	QFN-48-1EP_7x7mm_P0.5mm	1	U2
QFP48_SRAM	LQFP-48_7x7mm_P0.5mm	1	U5

DRC Status

Metric	Count
Errors	0
Warnings	0
Blocking	0

Status: PASS

Manufacturing Readiness

Verdict: WARNING

Action Items

- **[OPTIONAL]** Verify zone fill in KiCad: 3 nets appear incomplete but may be connected via zone fills
- **[OPTIONAL]** Verify zone fill in KiCad for 3 zone-connected nets

Routing Status

Metric	Value
Signal Net Completion	90.3% (28/31)
Overall Completion	85.3%
Complete Nets	29 / 34
Zone-Connected Nets	3
Incomplete Nets	5
Unconnected Pads	161

Zone-Connected Nets

- +1V2
- +1V8
- GND

Unrouted Signal Nets

- DQS_P
- MIPI_DAT1_N
- TMDS_D0_N

Unrouted Signal Nets

- DQS_P
- MIPI_DAT1_N
- TMDS_D0_N

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~4.09 USD
Components (estimated)	~2.8 USD
Assembly (estimated)	~2.05 USD
Total (estimated)	~8.94 USD
Batch Quantity	5
Batch Total (estimated)	~44.72 USD